

Consider a distributed database of a Hospital. Figure – 1 provides connection between different sites and indicates the local DBMSs. A global relation **Hospital** is fragmented (see Figure – 2) and allocated to different sites to create physical images. Figure – 3 shows the allocation schema. There are two applications A and B executing at site 1 and 4 respectively. Figure – 4 represents the implementations of these applications.

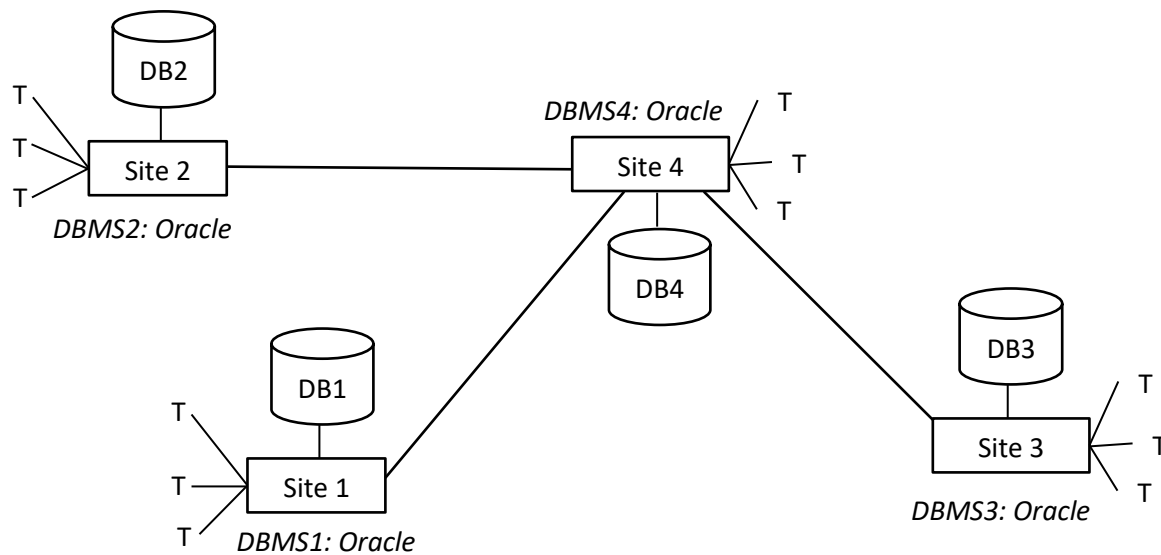


Figure – 1

Global Schema:

Hospital (HNAME, HID, CITY, MGRID, CAPACITY, CHARGE, RATINGS)

Fragmentation Schema:

$Hospital_1 = PJ_{HNAME, HID, CITY, MGRID} (Hospital)$

$Hospital_2 = SL_{CAPACITY < 500} PJ_{HID, CAPACITY, CHARGE, RATINGS} (Hospital)$

$Hospital_3 = SL_{RATINGS = 0} SL_{CAPACITY \geq 500} PJ_{HID, CAPACITY, CHARGE, RATINGS} (Hospital)$

$Hospital_4 = SL_{RATINGS > 0} SL_{CAPACITY \geq 500} PJ_{HID, CAPACITY, CHARGE, RATINGS} (Hospital)$

Figure – 2

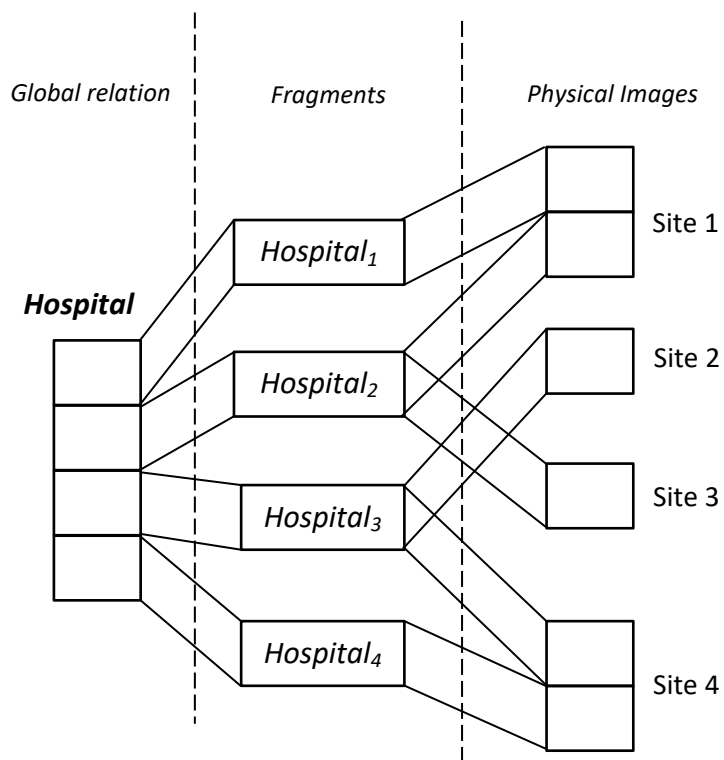


Figure – 3

Application – A:

Select * from ***Hospital₃***²;

Application – B:

Select * from ***Hospital₄***⁴;

Figure – 4

Now answer the following question.

1. [1 mark] What type of network topology is applied here?
2. [2 mark] What type of distributed database management system is this? Show reason behind your answer.
3. [2 marks] Which application is global? Show reason behind your answer.
4. [4 marks] Write the reconstruction formula to obtain *Hospital* from its fragments.
5. [5 marks] Discuss completeness property of the fragments of *Hospital*.
6. [6 marks] Discuss the Level-2 distribution transparency based on the given scenario.